A Unified Science of Perceptual/Motor Control through Sensor-Grounded Ontology

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

There are only two scientific questions - What am I? ...and what is all of *that*?

To answer the latter, we build massive space telescopes into the slopes of dark mountains and microscopes in the deep basements of tall buildings.

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Figure 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions.

DRAFT - WORK IN PROGRESS

1. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [1–3]
  - etc
  - Ferret work
    -

2. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

2.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending 'mesoscale research revolution' to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

3. Senior/Key Personnel Qualifications

3.1. Jonathan Matthis, PhD [Principle Investigator]

3.2. EI

3.3. AC,PhD

3.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President/CEO	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics[1–3]
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship

Table 2: Senior/Key Personnel Qualifications

4. Team Capabilities Statement

4.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

4.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

4.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

4.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

Bibliography

1. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28,** 1224–1233.e5 (2018).
2. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18,** e1009575 (2022).
3. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12,** e82410 (2023).